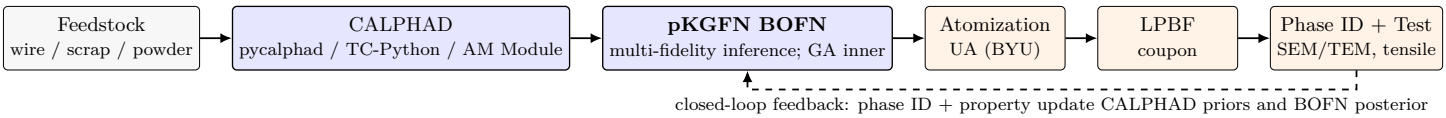
**DMREF Concept Outline — Critical-Mineral-Lean Printable Aluminum Alloys via CALPHAD–BOFN Discovery and Ultrasonic Atomization**

**PI:** Sterling Baird (Brigham Young University, Materials Science & Engineering). **Solicitation:** NSF DMREF (NSF 25-508). **Type:** 4-yr, multi-PI, ∼$1.5–2.0M. **Anticipated cycle:** Jan–Feb 2027. **Keywords:** MGI, BOFN, CALPHAD, LPBF, ultrasonic atomization, critical materials.




**Fundamental question**

What controls the printability and elevated-temperature phase stability of dilute multi-principal-element Al alloys built from Mn/Cr/Zr/Si/Fe additions, in the regime where conventional CALPHAD assessments are sparse and rapid-solidification AM thermal histories drive non-equilibrium phase selection? The MGI-relevant gap is not optimization of an existing alloy family but predictive thermodynamic understanding of a region of Al-MPEA phase space that has been opaque because the validation pipeline (CALPHAD → atomization → AM → phase ID) has never been closed end-to-end.

**Idea**

A 4-yr DMREF project using a single joint pKGFN BOFN acquisition over a constrained Al-MPEA design space (Al + dilute Mn/Cr/Zr/Si/Fe, ≤6 active solute elements, with the remaining critical-materials-relevant elements held at zero unless evidence supports activation), with CALPHAD multi-fidelity nodes (`pycalphad`, TC-Python/TCAL10, Thermo-Calc AM Module) as parallel partial-evaluation/context nodes. UA + LPBF serve as the experimental validation tier; their parameters are nuisance variables, not design targets. **Hypothesis:** the rapid-solidification phase fields accessible via UA + LPBF in dilute Al-Mn-Cr-Zr-Si space contain metastable strengthening phases that are absent from current CALPHAD databases and that can be predicted by a closed BOFN–CALPHAD–experiment loop. The scientific deliverable is a measured, openly published correction to the Al-MPEA phase diagram in this regime, with the BOFN–CALPHAD loop as the methodological tool. Composition/process/property records are deposited to the Materials Data Facility with cradle-to-coupon provenance.

**Why DMREF (MGI fit, multi-PI structure)**

*MGI alignment:* closed-loop computation + experiment + open data targeting fundamental thermodynamics, not parameter tuning of a known alloy. *Multi-PI structure (NSF-fundable):* BYU (Baird, lead — BOFN, CALPHAD, phase ID, tensile). *Unfunded U.S. collaborator:* U. Utah (Spear lab — no-cost LPBF coupon access via established MOU; not a subaward). *Unfunded international collaborator:* U. Toronto (J. Hattrick-Simpers, Acceleration Consortium — autonomous-experimentation methods cross-validation; supported through separate NSERC funds, not via NSF subaward). *Unfunded national-lab collaborator:* CMI Hub at Ames (existing letter; supply-risk context only; per DMREF rules Ames brings its own DOE funding). *Industrial collaborator:* Thermo-Calc (CALPHAD database access).

**Preliminary results (FLAIME pipeline, BYU, Apr 2026) and Y1 scientific milestone**

AMAZEMET rePOWDER UA atomizer installed and operational; UA composition modification from AlSi10Mg + Al wire + Si feed confirmed by SEM/EDS (the wire-to-powder pathway is in working form). AlSi10Mg LPBF tensile coupon printed at U. Utah (Spear lab); elevated-temperature tensile data on a standard Al specimen. Working pKGFN BOFN implementation; Thermo-Calc AM Module simulations on candidate FLAIME compositions; CALPHAD jobs on BYU HPC. *Y1 scientific milestone (the prediction-validation example DMREF reviewers should expect):* one closed-loop case in which the BOFN–CALPHAD posterior predicts a metastable phase in a defined Al-Mn-Cr-Zr-Si window that the public Al-database equilibrium assessment ranks differently, with SEM/TEM phase ID on a UA + LPBF coupon to confirm or refute the prediction.

**Scope separation from pending DOE Genesis Mission Topic 3B**

The two efforts are separated at the task and deliverable level, not just by narrative. *Genesis tasks (DOE-funded):* hardware/software integration of an autonomous LPBF-AM platform; scheduler; data fabric; alloy-agnostic platform validation. *DMREF tasks (NSF-funded):* (T1) hypothesis-driven CALPHAD-BOFN study of non-equilibrium phase selection in dilute Al-Mn-Cr-Zr-Si space; (T2) thermodynamic-database correction in that regime; (T3) open scientific data products (MDF deposition, CALPHAD overlay files). Personnel effort, deliverables, and reporting artifacts are itemized separately; no budget line, equipment purchase, or scheduler-development task appears in both. Current-and-pending will reflect this split explicitly.

**Request**

A brief read on whether this concept appears potentially responsive to DMREF and whether a short scoping call would be useful before BYU's internal limited-submission process for the Jan–Feb 2027 window.

*Repository and full vision:* https://github.com/vertical-cloud-lab.